DE–1710 **101**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2015.

First Semester

DIGITAL COMPUTER ORGANIZATION

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Use 2's complement to subtract the following :
   (a) $100101_2 - 1111_2$ (b) $11101_2 - 11110_2$
   (c) $0111_2 - 010_2$ (d) $1010_2 - 0101_2$.

2. Prove the theorem of Boolean algebra $X + 1 = 1$.

3. Explain the difference between a decoder and a demultiplexer.

4. What is an XOR gate? Write its truth-table and realize it using AND, OR and NOR gates.

5. Explain ring counter with a neat diagram.

6. Construct logic diagram for the Boolean expression $xyz + \bar{x}y\bar{z} + \bar{y}z + x\bar{y}z$.

7. Explain in detail about programmed controlled I/O.

8. Draw the circuit of a shift register and explain its functions in detail.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Simplify the expression using K – Map $F(A, B, C, D, E) = \sum (0, 1, 2, 4, 7, 8, 12, 14, 15, 16, 17, 18, 20, 24, 28, 30, 31)$.

10. Simplify the following expressions :

    (a) $AB + AB' + A'C + A'C'$

    (b) $xy + xyz' + xyz' + xzy$.

11. Explain the updown parallel counter with necessary diagram.

12. Explain, in detail the instruction cycles.

13. What do you mean by interrupt? What are its types? Why are they used? Explain, what happens when a computer recongnizes an interrupt?

14. Explain J-K master slave Flip-Flop with neat diagram.

15. (a) Write the functions of I/O interface.

    (b) Explain DMA in detail.

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DE–1711 **102**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2015.

First Semester

C AND DATA STRUCTURES

(2010 academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the various data types available in C.

2. Write a short note on macros.

3. Explain the various decision making statements available in C.

4. Define array. Describe multidimensional array with example.

5. Briefly explain the structures in C.

6. How do you pass pointers to function? Explain.

7. Describe the stack data structure.

8. Explain the singly linked list with example.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the 'gets' and 'puts' functions with example.

10. Explain the various looping statements available in C.

11. Define function. How do you passing arguments to a function? Explain.

12. Write a program in C to perform multiplication of two matrices.

13. Discuss the pointers in C with example.

14. Define tree. List the applications of tree.

15. Discuss the queue data structure and its representation.

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DE–1712 **103**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2015.

First Semester

RDBMS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Define primary key, foreign key and super key.

2. List the advantages of RDBMS over traditional file management system.

3. How data can be manipulated in SQL? Explain.

4. Define view. How it can be created in PLSQL?

5. What are the various integrity constraints supported by oracle?

6. What is a data base model? Discuss the advantages of relational model.

7. What is a trigger? How to invoke a trigger on demand?

8. List and explain the commands available in DML.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions.

9. Discuss the advantages of implementing RDBMS system in an organization.

10. Explain with suitable example about selecting distinct values using sub queries and union.

11. Describe the basic features of the SQL statement. List the advantages and disadvantages of the SQL statement.

12. Describe the three levels of data abstraction.

13. Explain the SQL commands for data definition language and data manipulation language.

14. Explain the following commands.

    (a) Commit

    (b) Roll back

    (c) Save point.

15. Define foreign keys. How does it play a role in the join operation?

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DE–1713 **104**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2015.

First Semester

OPERATING SYSTEMS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Differentiate the tightly coupled system from loosely coupled system.

2. Discuss the essential properties of the following types of operating system.

   (a) Time sharing

   (b) Handheld.

3. Explain the context switching with example.

4. What are the characteristics of deadlock? Explain.

5. Distinguish between the paging and segmentation.

6. Explain the swapping in detail.

7. What are the file protection mechanisms in file management? Explain.

8. Describe the basic file operations.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain in detail the types of system calls provided by a typical operating system.

10. Define process. Explain about process states and process control block with neat sketch.

11. Consider the following set of processes, with the length of the CPU burst given in milli seconds :

Process	Burst time	Priority
P1	10	3
P2	1	1
P3	2	3
P4	1	4
P5	5	2

The processes are assumed to have arrived in the order P1, P2, P3, P4, P5 all at time 0.

(a) Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms : FCFS, SJF, non preemptive priority (a smaller priority number implies a higher priority), and RR(quantum=1).

(b) What is the turnaround time of each process for each of the scheduling algorithms in part a?

(c) What is the waiting time of each process for each of the scheduling algorithms in part a?

(d) Which of the algorithms in part a results in the minimum average waiting time (overall processes)?

12. What are the goals of I/O software? Explain in detail.

13. Briefly explain the demand paging techniques in detail.

14. Describe in detail the types of directory structure.

15. Discuss the operating system structure with neat diagram.

DE–1714 **201**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2015.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the various benefits of Oops.

2. What are the derived data types available in C++? Explain them with example.

3. Briefly explain the uses of memory management operators.

4. Explain the working principles of 'for loop' with an example.

5. What is inline function? Explain with an example.

6. What are the points to be considered when constructing a constructor? Give an example.

7. How will you manipulate strings using operator overloading?

8. Explain the opening and closing of a file.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. What are the different types of operators available in C++? Explain them with an example.

10. (a) Explain the syntax of 'switch statement', and give an example. (8)

    (b) What is pointer? What are its benefits? (7)

11. Explain the 'friend function' with suitable example.

12. Explain the overloading of uniary and binary operators.

13. What are the different types of inheritance? Explain them with suitable example.

14. Explain the use of pointers to objects and pointers to arrays with an example.

15. Write a note on :

    (a) Command line arguments

    (b) File pointers.

DE–1715 **202**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2015.

Second Semester

UNIX AND SHELL PROGRAMMING

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write the features of UNIX operating system.

2. Write short note on I–node structure.

3. Explain the command line structure in detail.

4. Describe the shell variables.

5. Explain the various conditional statements available in shell programming.

6. Describe the 'zap' and 'pick' commands.

7. Discuss the file comparison in shell programming with various option.

8. Describe the control flow and relational operators.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Briefly discuss about files and directories structure of Unix operating system.

10. Explain the grep family of commands through examples.

11. Write a shell program to find the greatest of 3 numbers, input, as the command line parameter.

12. Explain the following built-in function with suitable example.

    (a) System ( )

    (b) Wait ( )

    (c) Stat ( ).

13. How to execute command without invoking a new process.

14. List the features of TBL and EGA preprocessors.

15. (a) Describe the bugs debugging in Unix. (8)

    (b) Write a short note on Ms–macro package. (7)

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DE–1716 **203**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2015.

Second Semester

COMMUNICATION SKILLS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Trace different types and modes of communication.

2. What are the barriers of communication?

3. Write a note on mack interview.

4. How is body language a powerful communication?

5. Differentiate upward and downward communication.

6. What are the objectives of negotiation techniques?

7. Explain the art of guiding.

8. Give an account on communication in management.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the concepts and definitions of communication.

10. Explain the importance of listening and speaking skills.

11. Bring out the salient features of group discussion.

12. Illustrate the Negotiation techniques.

13. Analyse different types of presentation.

14. Distinguish communication within management and communication in management.

15. Describe the process of group discussion.

DE–1717 **204**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2015.

Second Semester

COMPUTER GRAPHICS

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Differentiate between random scan and raster scan systems with suitable examples.

2. Write down the properties of a line.

3. Describe two – dimensional composite matrix transformation.

4. Write a short note on principles of transformation.

5. Explain the basic three dimensional transformation.

6. Describe any three 3-D display techniques.

7. Discuss the Depth-buffer algorithm.

8. Explain menu – driven command language.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the circle – generation algorithm with suitable example.

10. What is tablets? Explain the various types of tablet device.

11. Explain the midpoint subdivision line clipping algorithm.

12. What is transformation? Explain the basic 3–D transformation.

13. Explain the three dimensional clipping process.

14. What is coherence? Explain the scan-line coherence algorithm.

15. Describe the command language design issues.

DE–1718 **301**

DISTANCE EDUCATION

M.C.A./M.C.A.(Lateral) DEGREE EXAMINATION, MAY 2015.

Third Semester

DISCRETE MATHEMATICS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Show that $(P \to Q) \wedge (R \to Q) \Leftrightarrow (P \vee R) \to Q$.

2. Show that $I_{12} : \rceil Q, P \to Q \Rightarrow \rceil P$.

3. Define power set. Write the power set of $\{a, \{b\}\}$.

4. If $X = \{1,2,...7\}$ and if $R = \{< x, y > / x - y$ is divisible by 3}. Show that R is an equivalence relation.

5. Write a note on hashing function with an example.

6. For any commutative monoid $< M, * >$, prove that the set of idempotent elements of M forms a submonoid.

7. Show that the set of all elements a of a group $< G, * >$, such that $a * x = x * a$ for every $x \in G$ is a subgroup of G.

8. Define the following with an example

(a) Adjacency matrix of a graph

(b) Directed tree

(c) Weighted graph

(d) Path of the graph.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Obtain the principal conjuctive normal form of the formula S given by $(\neg P \to R) \wedge (Q \rightleftarrows P)$

(b) Obtain the principal disjuctive normal form of the formula $P \vee (\neg P \to (Q \vee (\neg Q \to R)))$. (8 +7)

10. (a) Show that $(x)(P(x) \vee Q(x)) \Rightarrow (x)P(x) \vee (\exists x)Q(x)$.

(b) Construct the derivation of the equivalence $P \to (\exists x)Q(x) \Leftrightarrow (\exists x)(P \to Q(x))$ without using E_{32}.

(8 + 7)

11. (a) If relations R and S are reflexive, symmetric and transitive show that $R \cap S$ is also reflexive symmetric and transitive.

(b) If A be a given finite set and $\rho(A)$ its power set. If $\subseteq$ be the inclusion relation on the elements of $\rho(A)$. Draw Hasse diagrams of $< \rho(A), \subseteq >$ for (i) $A = \{a\}$ (ii) $A = \{a,b\}$ (iii) $A = \{a,b,c\}$. (8 + 7)

12. (a) If $X = \{1,2,3\}$ and f, g, h and s be functions from X to X given by $f = \{<1,2>, <2,3>, <3,1>\}$
$g = \{<1,2>, <2,1>, <3,3>\}$
$h = \{<1,1>, <2,2>, <3,1>\}$
$s = \{<1,1>, <2,2>, <3,3>\}$ find $f \circ g$; $g \circ f$; $f \circ h \circ g$
$s \circ g$; $g \circ s$; $s \circ s$ and $f \circ s$.

(b) Show that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. (8 +7)

13. (a) Prove that a subset $S \neq \phi$ of G is a subgroup of $< G, * >$ iff for any pair of elements $a, b \in S; a * b^{-1} \in S$.

(b) Prove that every finite group of order n is isomorphic to a permutation group of degree n. (8 +7)

14. (a) Prove that the order of a subgroup of a finite group divides the order of the group.

(b) If $< G, * >$ and $< H, \Delta >$ be groups and $g : G \to H$ be a homomorphism. Then prove that the kernel of g is a normal subgroup. (8 + 7)

15. (a) Discuss in detail about the matrix represention of graphs.

(b) Explain different types of trees with example. (8 + 7)

DE–1719 **302**

DISTANCE EDUCATION

M.C.A. /M.C.A (Lateral) DEGREE EXAMINATION,
MAY 2015.

Third Semester

COMPUTER NETWORKS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write a note on network hardware.

2. Write short notes on geostationary satellites.

3. Explain the simplex stop-and-wait protocol with an example.

4. Discuss any one of the algorithm which allocate multiple access protocols.

5. Explain shortest path routing with an example.

6. What are the elements of transport protocols? Discuss flow control and buffering.

7. Explain cipher modes in symmetric key algorithm.

8. Write a note on compression.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the TCP/IP reference model and compare with OSI model.

10. Explain error-correcting and detecting codes with examples.

11. Discuss carrier sense multiple access protocols in detail.

12. What is routing algorithm? Discuss any two routing algorithm with examples.

13. Write short notes on :

    (a) TCP service model

    (b) TCP transmission policy.

14. Discuss public-key algorithm in detail.

15. Explain audio and video compression with examples.

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DE–1720 **303**

DISTANCE EDUCATION

M.C.A. (Sem)/ M.C.A. (Lateral)
DEGREE EXAMINATION, MAY 2015.

Third Semester

SOFTWARE ENGINEERING

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Brief on Software process and size factors.
2. Discuss on Reusable Software Resources.
3. What are the basic principles of Project scheduling?
4. Explain version control in Software Configuration Management.
5. Brief on Requirements Specification.
6. Discuss on Use-Cases.
7. What is Software Design process?
8. Discuss on White Box testing.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Detail on Evolutionary Software Process Models.
10. Explain the following :
    (a) Human Resources. (5)
    (b) Software sizing. (5)
    (c) Software scope. (5)

11. Explain in detail about Risk Identification and Risk Projection.

12. Describe the Software Quality concepts and Quality Assurance.

13. Explain the Software Requirements Analysis and principles.

14. Detail on User Interface Design rules, models and activities.

15. Explain the Integration Testing strategies.

DE–1721 **304**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2015.

Third Semester

VISUAL PROGRAMMING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. List and explain the types of GDI objects.

2. What are the various user events in visual programming?

3. Explain the properties of DAO control.

4. Write short notes on serialization.

5. Explain mode and modeless dialog boxes.

6. Write a short note on splitter window.

7. Define SDI. Explain briefly.

8. Discuss about the features of visual basic and visual C++.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. List out the various statements in visual basic and explain in detail.

10. Discuss the file system controls in detail.

11. What are the visual C++ components? Explain any three components in detail.

12. Discuss in detail the dialog based applications.

13. Define DLL. Explain the process of creating user defined DLL's.

14. Describe the dynamic data transfer function.

15. Explain the user interface classes in detail.

DE–1722 **401**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2015.

Fourth Semester

ACCOUNTING AND FINANCIAL MANAGEMENT

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. The following Trial Balance has been prepared wrongly. You are asked to prepare the Trial Balance correctly.

Head of accounts	Dr. Bala	Cr. Bala
	Rs.	Rs.
Capital	22,000	–
Stock	–	10,000
Debtors	8,000	–
Creditors	–	12,000
Machinery	–	20,000
Cash in hand	–	2,000
Bank overdraft	14,000	–
Sales returns	–	8,000
Purchase returns	4,000	–
Misc. expenses	12,000	–
Sales	–	44,000

Head of accounts	Dr. Bala Rs.	Cr. Bala Rs.
Purchases	26,000	–
Wages	10,000	–
Salaries	–	12,000
Prepaid insurance	–	200
Bills payable	10,800	–
Outstanding Salaries	1,400	–
	1,08,200	1,08,200

2. What are the different types of accounts? State their rules.

3. From the following information find out

   (a) Current assets

   (b) Current liability

   (c) Value of inventory

      (i) Current ratio = 3.5

      (ii) Liquid ratio = 2.5

      (iii) Working capital = Rs. 1,00,000.

4. What are the uses of ratio analysis?

5. Calculate the material, (a) cost variance (b) price variance (c) quantity variance.

	Standard			Actual		
	Qty	Rate Rs.	Amount Rs.	Qty	Rate Rs.	Amount Rs.
A	4	100	400	2	350	700
B	2	200	400	1	200	200
C	2	400	800	3	300	900
	8		1,600	6		1,800

6. Prepare a production budget from the following information.

Product	Estimated stock on 1.1.2007 (units)	Estimated sales during Jan. to March 2007 (units)	Desired closing stock on 31.3.2007 (Units)
R	2,000	10,000	3,000
S	3,000	15,000	5,000
U	4,000	13,000	3,000
P	3,000	12,000	2,000

7. Explain the merits and demerits of standard costing.

8. Ezhil Ltd. is producing articles mostly by manual labour and is considering to replace it by a new machine. There are two alternative models P and R of the new machine. Prepare a statement of profitability showing the pay-back period from the following information.

	Machine P	Machine R
Estimated life of machine	4 years	5 years
	Rs.	Rs.
Cost of machine	9,000	18,000
Estimated savings in scrap	500	800
Estimated savings in direct wages	6,000	8,000
Additional cost of maintenance	800	1,000
Additional cost of supervision	1,200	1,800

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. From the following Trial balance has been extracted from the book of Thiru. Velusamy on 31.12.2008. From these particulars prepare Trading and profit and loss a/c for the year ending 31.12.2008 and also balance sheet as on that date.

Trial Balance

Particulars	Rs.	Particulars	Rs.
Machinery	4,000	Capital	9,000
Cash in hand	500	Sales	12,000
Cash at bank	1,000	Bank loan	4,000
Wages	1,000	Sundry creditors	4,500
Purchases	8,000	Dividend received	300
Stock 1.1.2008	6,000		
Sundry debtors	4,400		
Bills receivable	2,900		
Rent	400		
Interest on bank loan	50		
Commission	250		
General expenses	800		
Salaries	500		
	29,800		29,800

Adjustments :

(a) Closing stock 31.12.2008 Rs. 8,000

(b) Wages outstanding Rs. 100

(c) Salaries unpaid Rs. 100

(d) Rent prepaid Rs. 150

(e) Commission Rs. 50

(f) Interest on bank loans not paid Rs. 400.

10. Following are the summarised Balance sheets of A & Co Ltd. as on 31st December 1999 and 2000.

Balance Sheet

Liabilities	1999 Rs.	2000 Rs.	Assets	1999 Rs.	2000 Rs.
Share capital	1,00,000	1,50,000	Land and buildings	1,00,000	90,000
General reserve	50,000	60,000	Plant and machinery	1,00,000	1,19,000
Profit and loss a/c	30,500	30,000	Stock	50,000	24,000
Bank loan	70,000	–	Debtors	75,000	63,000
Sundry creditors	50,000	37,200	Cash	500	1,200
Provision for tax	32,000	35,000	Bank	2,000	15,000
			Goodwill	5,000	–
	3,32,500	3,12,200		3,32,500	3,12,200

Additional information :

(a) Dividend of Rs. 23,000 was paid during 2,000

(b) Depreciation written off on building Rs. 10,000 and machinery Rs. 14,000

(c) Income tax paid during the year ended 31st December 2000 Rs. 28,000

Prepare a funds flow statement for 2000.

11. Explain the various methods of evaluating capital expenditure.

12. A company shows the following results for two periods.

Period	Sales	Profit
I period	Rs. 20,000	Rs. 1,000
II period	Rs. 10,000	Rs. 400

Calculate :

(a) P/V ratio

(b) Fixed cost

(c) BEP

(d) Profit when sales are Rs. 30,000.

13. For production of 10,000 electrical automatic irons the following are budgeted expenses.

	Per unit Rs.
Direct materials	60
Direct labour	30
Variable overheads	25
Fixed overheads (Rs. 1,50,000)	15
Variable expenses (direct)	5
Selling expenses (10% fixed)	15
Administration expenses (Rs. 50,000 rigid for all levels of production)	5
Distribution expenses (20% fixed)	5
Total cost of sale per unit	160

Prepare a budget for production of Rs. 6,000, 7,000 irons showing distinctly marginal cost and total cost.

14. Write short notes on :

(a) Zero base budget

(b) Master budget

(c) Production budget.

15. You are required to prepare a statement of working capital needed to finance a level of activity of 5,200 units of output, you are given the following information.

Elements of cost	Amount per unit Rs.
Raw materials	8
Direct labour	2
Overheads	6
Total cost	16
Profit	4
Selling price	20

Raw materials are in stock on an average one month. Materials are in process on an average-half a month. Finished goods are in stock on an average-6 weeks. Credit allowed by creditors is one month. Lag in payment of wages is 1.5 weeks cash in hand and at bank is expected to be Rs. 7,300, credit allowed to debtors is two months.

You are informed that production is carried on evenly during the year and wages and overheads accure similarly.

DE–1723 **402**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2015.

Fourth Semester

INTERNET PROGRAMMING

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write a note on common gateway interface.

2. Differentiate Internet and Intranet applications.

3. How do you create objects and define functions in JavaScript? Explain.

4. List any eight built-in functions available in VBScript.

5. Explain the following tags in HTML

   (a) Ordered list

   (b) Unordered list

   (c) Table

   (d) Frame.

6. Briefly explain the VDO live technology.

7. Write a program to create an ActiveX control to activate a web page.

8. How do you insert images into a web page and how do you make it as hyperlink? Explain.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the function overloading and overriding concepts in Java.

10. Describe the design issues in WWW.

11. Define event. Explain the various event handlers available in JavaScript.

12. What do you mean by client-side and server-side validation? How do you perform client-slide validation using VBScript? Explain.

13. Write a note on Shockwave and Lingo.

14. How do you create a custom integrated application with multiple protocols? Explain.

15. Explain the dynamic graphics and animation.

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DE–1724 **403**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2015.

Fourth Semester

OBJECT ORIENTED ANALYSIS AND DESIGN

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Describe the issues related to the object model.

2. What are the common approaches have evolved in programming languages to capture generalisation?

3. Explain the components of architecture modeling.

4. What are the suggestions to be follows in refining key abstraction? Explain.

5. Explain the activity diagram with an example.

6. Describe the basic activities of an abstractor.

7. How case modeling is used to describe the functional requirements? Explain with example.

8. Describe the strategies used to identify the conceptual classes.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Briefly explain the various phases related to identify the classes and objects.

10. Explain the different phases of unified process.

11. Discuss any two general approaches in classification.

12. Write notes on :

    (a) Refactoring

    (b) Key abstraction mechanism.

13. Briefly explain the various classes of modeling.

14. Discuss the UML deployment and component diagram, in detail.

15. Explain the design patterns and the principles used in analysis and design.

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DE–1725 **404**

DISTANCE EDUCATION

M.C.A. (S) / M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2015.

Fourth Semester

COMPILER DESIGN

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write a note on lexical analyzer.

2. Define transition diagram and its notations.

3. What is parse tree? Give the characteristics of the parse tree.

4. Write short notes on LALR parsing.

5. Briefly explain the three address code.

6. What is symbol table? List the contents of symbol table.

7. List out the applications of DAG.

8. Write a note on Peephole optimization.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Construct a minimum state DFA for the Regular Expression (a/b) (a/b) (a/b).

10. Explain the implementation of a lexical analyzer.

11. Explain the method of constructing the operator precedence functions.

12. Discuss shift reduce parsing with suitable example.

13. Explain synthesized attributes with an example.

14. Explain the terms in detail :

    (a) Syntatic errors. (9)

    (b) Semantic errors. (6)

15. Explain in detail about code generation algorithm.

DE–1726 **501**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2015.

Fifth Semester

DISTRIBUTED COMPUTING

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Describe various Internet protocols used for distributed systems.

2. Explain about two phase commit protocol for nested transactions.

3. Explain in detail

   (a) CORBA's CDR

   (b) XML.

4. Explain the file service architecture for distributed systems.

5. Explain DNS with an example.

6. Explain

   (a) WWANS (b) Internet works

   (c) Data stream (d) Congestion control

7. Discuss layers in Distributed Operating System.

8. Explain

   (a) Primary Backup Replication

   (b) Active Replication.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. What is Architectural model? Explain in detail the architectural element and architectural Pattern.

10. Explain with an example, how to implement group communication and client server communication.

11. Discuss election algorithm in detail.

12. Explain the execution environment in distributed operating system.

13. Discuss the 'snapshot' algorithm of Chandy and Lamport.

14. Explain digital signature in detail.

15. Explain transaction recovery in distributed system.

DE–1727 **502**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2015.

Fifth Semester

WEB TECHNOLOGY

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What is the use of table? List out the table tags in HTML.

2. Write a Java Script program to find largest among five numbers.

3. Define BDK. Write the steps to create Beanbox.

4. List out the bean properties and its characteristics.

5. How do you execute a program using servlet? Explain with an example.

6. Write a note on JSP environment.

7. What is JSTL? List out the core tag in JSTL.

8. How to delete a cookie within a JSP? Explain.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the concept of CSS and its uses with an example.

10. Explain in detail about XML. Give an example.

11. How to generate applets using the Bean Box? Explain.

12. Describe the HTTP GET and Post request in servlets.

13. Discuss JSP Technology in detail.

14. Discuss about the error handling and debugging in JSP pages.

15. Explain the steps to connect database using JDBC.

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DE–1728 **503**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2015.

Fifth Semester

.NET FRAME WORKS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What happens when you compile and run a program in .NET framework? Explain.

2. List the advantages of .NET framework.

3. Explain any five string handling functions available in .NET.

4. Compare and contrast .NET framework with other programming languages.

5. Define thread. Explain the various states of thread with an example.

6. Describe any five controls available in .NET.

7. Write a short note on events and delegates.

8. Briefly explain the WSDL.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the architecture of .NET framework.

10. Discuss the array list and collections in detail.

11. Write a detailed note on synchronization.

12. Describe the DataGrid control in .NET framework.

13. How do you expose web-services using .NET framework? Explain.

14. Explain the ADO.NET with example.

15. Explain the various looping statements available in .NET framework.

DE–1729 **504**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A.(S)(Lateral) DEGREE EXAMINATION, MAY 2015.

Fifth Semester

DATA MINING AND WAREHOUSING

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. How do you integrate the data mining system with a data warehouse system?
2. Distinguish between data mining and data warehousing.
3. What is KDD? Explain.
4. Mention the features of data warehouse.
5. List the data mining applications.
6. What are the performance enhancement techniques available in data warehouse?
7. List any five security policy in data warehousing.
8. How do the statistical techniques help in fine tuning of the data warehouse?

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the various data mining task primitives in detail.
10. What are the major issues in data mining? Explain.

11. What are the three major types of metadata in a data warehouse? Briefly mention the purpose of each type.

12. What are the differences between operational database systems and data warehouses?

13. Explain, in detail, the architecture of data warehouse.

14. Give reasons, why the data warehouse must be backed up? State the features of datawarehouse.

15. Explain the different methods are available for testing the data warehouse.

DE–1730 **601**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2015

Sixth Semester

MOBILE COMMUNICATIONS

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What are the different kinds of mobility? Explain with suitable examples.

2. Why does CSMA/CD fail in wireless network? Reason out.

3. List out the services provided by GSM technology and explain two of them with suitable examples.

4. What are Ad-Hoc Networks? Explain briefly.

5. What is meant by Roaming? Bring out the steps involved in roaming between access points.

6. Explain how does Mobile IP serve as a standard to enable mobility in the Internet.

7. Describe the mechanism used for forwarding packets.

8. What is I-TCP? Why is it necessary? List out its advantages.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Name the main elements of the GSM system architecture and describe their Functions with a neat sketch.

10. (a) What advantages does the use of IPv6 offer for mobility? Where are the entities of mobile IP? Explain

(b) What are general problems of mobile IP regarding security and support of Quality of service? Explain.

11. Discuss the various reasons led to the development of WATM and also explain where WATM can be used.

12. Compare IEEE 802.11, HiperLAN2, and Bluetooth with regard to their ad-hoc capabilities. Where is the focus of these technologies? Explain.

13. Explain the various mechanisms of the transmission control protocol that influence the efficiency of TCP in a mobile environment.

14. Compare the different types of transmission errors that can occur in wireless and wired networks.

15. Write Short Notes on :

(a) WAP protocol

(b) WML.

DE–1731 **602**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2015.

Sixth Semester

MIDDLEWARE TECHNOLOGY

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write a note on J2EE architecture.

2. Explain the salient features of 3-tier client-server architecture.

3. Explain the life cycle of servlets.

4. Describe the layer of architecture of CORBA.

5. What are the steps involved in a EJB conversation? Explain.

6. Explain the ASP-NET parser.

7. Write a note on WSDL.

8. Compare and contrast the ADO and ADO.NET.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Compare and contrast the 2 tier architecture and 3 tier architecture.

10. Discuss any five feature of IDL. Explain how IDL resolves the disputes between programming languages.

11. Explain the life cycle of an entity bean in detail.

12. Discuss about statefull and stateless bean with an example.

13. Explain the following in ASP.NET

    (a) Adrotator controls

    (b) Web server controls

    (c) Validation controls

14. Explain the ASP.NET object data source control.

15. Explain the use of UDDI and SOAP in web services.

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DE–1732 **603**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2015.

Sixth Semester

AGENT BASED INTELLIGENT SYSTEM

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. List out the various searching strategy.

2. Write down the concepts of first order logic.

3. Discuss the forward and backward chaining.

4. Describe the partial order planning approach with an example.

5. Explain about the fuzzy logic.

6. How symbolic reasoning is measured under uncertainty.

7. What is use of augmented grammer in natural language processing?

8. Discuss the reinforcement learning.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain in detail about the Heuristic Search techniques.

10. Discuss the different issues in knowledge representation.

11. How scripts and frames are defined and used for knowledge representation.

12. Discuss in detail about the genetic algorithm.

13. Explain the representation for planning in detail.

14. Explain the reasoning :

    (a) Temporal reasoning

    (b) Statistical reasoning.

15. Discuss the different stages involved in analyzing a sentence in natural language processing.

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DE–1733 **604**

DISTANCE EDUCATION

M.C.A. /M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2015.

Sixth Semester

OPEN SOURCE ARCHITECTURE

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Discuss FOSS.
2. Explain any five basic commands in Linux.
3. Write a note on X – window system.
4. Explain the file system process in Linux administration.
5. What is CVS? List some useful CVS commands.
6. Write a note on patch.
7. Give a brief note on Bash scripting.
8. What are the rules for local and global variables in Python?

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain services in Linux. How services are selected for startup in Linux?
10. How to install Linux operating in Bash? Explain with example.

11. Give a detailed note on various web servers.

12. Explain the process of setting a firewall using net finder and IP tables.

13. Explain the usage of CVS tool to manage source code revision.

14. Write in detail about GNU debugging tools.

15. Give a detailed note on the control statements used in Python programming with examples.